AI-900

AZURE AI

COURSE SLIDES · WSQ

Microsoft Azure AI Fundamentals (AI-900)

WSQ Course Code: TGS-2023021100

Trainer: Dr. Alfred Ang

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

- Confirm attendance, facilities, emergency procedures, and course timing.
- Use the LMS/TMS portal for courseware, digital attendance, feedback, and assessment.
- Keep this deck and Learner Guide available during class activities.

Digital Attendance (Mandatory)

- Take AM, PM, and assessment digital attendance when the QR code is displayed.
- Scan the QR code with your mobile phone camera and submit the attendance form.
- Minimum 75% attendance is required for funding eligibility.

YOUR TRAINER - GENERAL

About the Trainer

?

Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on AI agents, automation, cloud, data and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Azure, AI, analytics, or cloud services.
- One AI scenario you want to understand by the end of the course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

LMS / TMS

- Portal: https://lms-tms.tertiaryinfotech.com
- Download slides and Learner Guide from the portal.
- Courseware and approved materials may be used for the open-book assessment.

GITHUB COURSE REPOSITORY

Download the Labs

- Open: github.com/tertiarycourses/TGS-2023021100---Microsoft-Azure-AI-Fundamentals-AI-900-
- Complete the labs in order because later activities reuse the same Azure AI vocabulary.
- Use an instructor-provided Azure subscription, Microsoft Learn sandbox, or Azure free account.

DAY 1

Lesson Plan

Morning

- Digital attendance (AM)
- Trainer and learner introductions
- AI workloads and responsible AI
- Lab 01: AI workloads and responsible AI
- Lab 02: Machine learning fundamentals

Afternoon

- Digital attendance (PM)
- Azure Machine Learning and model lifecycle
- Lab 03: Azure Machine Learning
- Lab 04: Computer vision, image analysis, OCR
- Day 1 review

DAY 2

Lesson Plan

Morning

- Digital attendance (AM)
- NLP, speech, translation, document intelligence
- Labs 05-07: Azure AI services
- Generative AI, Foundry, OpenAI, model catalog

Afternoon

- Lab 08: Generative AI
- Lab 09: Solution design and governance
- Lab 10: Capstone and course review
- Digital attendance (Assessment) and final assessment
- TRAQOM survey and closing

WSQ TSC

Skills Framework

TSC Title

- Artificial Intelligence Application
- TSC Code: ICT-DIT-4016-1.1
- Cloud and AI service awareness for business and technical roles
- Responsible use of AI-enabled services

Applied Abilities

- Identify AI workloads from business requirements.
- Select suitable Azure AI services.
- Recognize governance, privacy, safety, and cost controls.

TSC Knowledge (K)

TSC Title: Artificial Intelligence Application | TSC Code: ICT-DIT-4016-1.1

K1	Artificial intelligence workload types and business use cases.
K2	Responsible AI principles, privacy, security, transparency, and accountability.
K3	Machine learning concepts including features, labels, training, evaluation, and deployment.
K4	Azure AI service capabilities for vision, language, speech, documents, and search.
K5	Generative AI concepts including prompts, foundation models, grounding, and content safety.
K6	Governance, access control, monitoring, cost, and resource cleanup considerations.

TSC Abilities (A)

Abilities are practiced through Labs 01-10 and referenced in the Lesson Plan slide numbers.

A1	Identify AI workloads from practical business scenarios and input/output requirements.
A2	Map Azure AI services to workload requirements and learner lab evidence.
A3	Apply responsible AI review questions to vision, language, speech, document, and generative AI use cases.
A4	Explain machine learning and generative AI concepts using simple workplace examples.
A5	Recommend basic security, privacy, monitoring, and cost controls for AI solutions.
A6	Prepare a capstone service map and review plan aligned to the labs.

SKILLS FRAMEWORK

Skills Framework links abilities to labs and assessment

Knowledge

- AI workloads
- Responsible AI
- Azure AI services
- Governance controls

>

Abilities

- Map scenarios
- Select services
- Validate outputs
- Explain risks

>

Practice

- Labs 01-10
- Checkpoints
- Capstone service map
- Cleanup review

>

Assessment

- Open-book MCQ
- Competency evidence
- Attendance
- TRAQOM

Alignment rule: every course outcome is practiced in the labs and referenced in the lesson plan slide numbers.

COURSE OUTCOMES

Learning Outcomes

- Identify common AI workloads and use cases.
- Explain responsible AI principles and human review needs.
- Distinguish regression, classification, clustering, deep learning, and transformer-based workloads.
- Describe Azure Machine Learning and automated ML concepts.
- Map vision, NLP, speech, translation, document, and generative AI scenarios to Azure services.
- Recognize security, privacy, governance, and cost controls for AI solutions.

TOPICS 1-4

Course Outline

Foundations

- AI workloads and responsible AI
- Machine learning fundamentals
- Features, labels, training, evaluation, deployment
- Azure Machine Learning and AutoML

Vision and Language

- Image analysis, OCR, object detection
- Sentiment, key phrase extraction, entity recognition
- Speech recognition, speech synthesis, translation

TOPICS 5-8

Course Outline

Documents and Generative AI

- Document Intelligence and knowledge mining
- Azure AI Search concepts
- Generative AI, prompts, LLMs, grounding, RAG
- Azure AI Foundry, Azure OpenAI, model catalog

Governance and Review

- Security, privacy, managed identity, RBAC
- Budget alerts and resource cleanup
- Capstone service mapping
- Assessment preparation and course review

ASSESSMENT

Final Assessment

- Format: written / MCQ assessment on the LMS.
- Assessment is open book: approved course slides, Learner Guide, and materials only.
- Complete the assessment digital attendance before starting.
- Appeals follow the centre's published assessment process.

Briefing for Assessment

- Place phones and other materials under the table or as instructed.
- No photos or recording of assessment scripts.
- No discussion during assessment.
- Submit answers on the LMS before the time ends.
- Sign the Assessment Summary Record when instructed.

Criteria for Funding

- Minimum attendance rate of 75% based on SSG digital attendance records.
- Complete the assessment and be assessed as Competent.
- Complete course feedback and TRAQOM survey when requested.

COURSE REPOSITORY

Access the Hands-On Labs

Use the GitHub repository to review lab files, revisit steps after class, and keep your Learner Guide aligned with the exercises.

https://github.com/tertiarycourses/TGS-2023021100---Microsoft-Azure-AI-Fundamentals-AI-900-

Option A

Clone the repository with git clone <course-repo-url>.git

Option B

Download a ZIP from GitHub: Code > Download ZIP

Run in class

Use Azure portal, Microsoft Learn, and instructor-provided accounts as directed.

BEFORE YOU GO

Certification & TRAQOM Survey

- Complete the certification and TRAQOM survey to receive your certificate.
- Confirm AM, PM, and assessment attendance are recorded.
- Keep the Learner Guide for post-course revision and workplace reference.

COURSE CONCEPTS

AI-900 builds recognition skill through scenarios

Learners practice mapping business needs to AI workloads, Azure services, and responsible controls.

IMPORTANT

Course Status Note

- Microsoft Learn states that Exam AI-900 retired on June 30, 2026.
- This courseware remains aligned to the AI-900 skills outline and Azure AI fundamentals concepts.
- The course should be positioned as Azure AI Fundamentals training, not as a guarantee of exam appointment availability.

CONCEPT

AI workload recognition is the core skill

- Prediction and classification: use data to estimate values or categories.
- Vision: interpret images, detect objects, and read text with OCR.
- Language and speech: understand text, translate, transcribe, and synthesize voice.
- Documents: extract structured information and make unstructured content searchable.
- Generative AI: draft, summarize, answer, and assist using foundation models.

CONCEPT

Azure AI services match workloads to capabilities

Start from the workload, then select the Azure AI capability that best fits the input, output, and control needs.

Machine Learning
Azure Machine Learning
Predict values, classify, cluster

Azure AI
Service selection hub

Speech & Translation
Azure AI Speech / Translator
Transcribe, synthesize, translate

Vision
Azure AI Vision
Analyze images, OCR, detect objects

Match scenario to service

Documents & Search
Document Intelligence / AI Search
Extract forms, index knowledge

Language
Azure AI Language
Sentiment, entities, key phrases

Generative AI
Azure AI Foundry / Azure OpenAI
Prompt, ground, review outputs

Lab connection: learners repeatedly identify the workload, choose the service, validate the result, and name a responsible AI control.

CONCEPT

Responsible AI keeps automation accountable

- Fairness: evaluate whether different groups are affected unequally.
- Reliability and safety: test outputs and define fallback paths.
- Privacy and security: protect data, access, keys, and logs.
- Inclusiveness: design for different languages, abilities, and contexts.
- Transparency and accountability: explain when AI is used and who owns decisions.

CONCEPT

Machine learning follows a repeatable lifecycle

- Define the prediction task and business success measure.
- Prepare features and labels from trustworthy training data.
- Train and evaluate a model using appropriate metrics.
- Deploy, monitor, retrain, and govern the model over time.

CONCEPT

Generative AI adds new controls

- Prompts guide the model, but outputs still need review.
- Grounding and retrieval reduce unsupported answers by adding source context.
- Content filters, evaluations, and monitoring help manage harmful or inaccurate outputs.
- Users should know when content is AI-assisted.

AI Workloads and Responsible AI

LAB 01

A retail company wants to use AI for product recommendations, customer support, document processing, image analysis, and marketing content generation. You must identify the workload types and responsible AI concerns.

Objectives

- Identify common AI workloads.
- Match workloads to business scenarios.
- Explain responsible AI principles.
- Create a responsible AI checklist.

Validation: You should have workload examples, Azure service mapping, and a responsible AI checklist.

LAB 1 - DIAGRAM

Lab 1 workflow diagram

1. 1. Identify AI Workloads
2. 2. Map Workloads to Azure Services
3. 3. Review Responsible AI Principles
4. 4. Build a Responsible AI Checklist

Lab evidence: You should have workload examples, Azure service mapping, and a responsible AI checklist.

LAB 1 - STEP FLOW

Lab 1 guided steps

- 1. Identify AI Workloads
- 2. Map Workloads to Azure Services
- 3. Review Responsible AI Principles
- 4. Build a Responsible AI Checklist

LAB 1 - REVIEW

Lab 1 checkpoint

Checkpoint questions

- What is a computer vision workload?
- Why is fairness important?
- What is transparency in AI?
- Who is accountable for an AI system?

Course focus

AI-900-style questions often ask you to identify workload types and responsible AI principles from a scenario.

Machine Learning Fundamentals

LAB 02

The retail company wants to use historical data to predict outcomes. You must decide which machine learning technique fits each business question.

Objectives

- Distinguish regression, classification, and clustering.
- Explain features, labels, training data, and validation data.
- Understand deep learning and transformer concepts.
- Match ML techniques to scenarios.

Validation: You should have scenario mappings, feature/label examples, and a training/validation explanation.

LAB 2 - DIAGRAM

Lab 2 workflow diagram

1 — 1. Classify ML Scenarios > 2 — 2. Identify Features and Labels > 3 — 3. Explain Training and Validation > 4 — 4. Review Deep Learning and Transformers > 5 — 5. Match Technique to Service

Lab evidence: You should have scenario mappings, feature/label examples, and a training/validation explanation.

Lab 2 guided steps

- 1. Classify ML Scenarios
- 2. Identify Features and Labels
- 3. Explain Training and Validation
- 4. Review Deep Learning and Transformers
- 5. Match Technique to Service

Lab 2 checkpoint

Checkpoint questions

- What is regression used for?
- What is classification used for?
- What is clustering used for?
- What is the difference between a feature and a label?

Course focus

Know how to select the basic ML technique from the wording of a business scenario.

LAB 3

Azure Machine Learning, Automated ML, Model Lifecycle

LAB 03

The retail company wants to build a custom churn model but does not yet have a large data science team. You must explain how Azure Machine Learning can support the workflow.

Objectives

- Describe Azure Machine Learning capabilities.
- Explain automated machine learning.
- Understand data, compute, jobs, models, and endpoints.
- Map the model lifecycle.

Validation: You should have a workspace area map, AutoML explanation, and model lifecycle flow.

Lab 3 workflow diagram

1. 1. Open Azure Machine Learning Studio
2. 2. Identify Core Workspace Areas
3. 3. Explain Automated ML
4. 4. Map the Model Lifecycle
5. 5. Choose Compute Safely

Lab evidence: You should have a workspace area map, AutoML explanation, and model lifecycle flow.

Lab 3 guided steps

- 1. Open Azure Machine Learning Studio
- 2. Identify Core Workspace Areas
- 3. Explain Automated ML
- 4. Map the Model Lifecycle
- 5. Choose Compute Safely

Lab 3 checkpoint

Checkpoint questions

- What is automated machine learning?
- What is a model endpoint?
- Why is model monitoring needed?
- Why does compute choice affect cost?

Course focus

Understand what Azure Machine Learning does at a high level: data, compute, training, automated ML, model management, deployment, and monitoring.

LAB 4

Computer Vision, Image Analysis, OCR

LAB 04

The retail company wants to analyze product photos, detect damaged items, read shelf labels, and blur faces in store images.

Objectives

- Identify computer vision workloads.
- Distinguish image classification, object detection, OCR, and face detection.
- Describe Azure AI Vision capabilities.
- Review responsible AI concerns for vision.

Validation: You should have a scenario mapping table, Azure AI Vision capability notes, and responsible AI review.

LAB 4 - DIAGRAM

Lab 4 workflow diagram

1. Match Vision Scenarios
2. Explore Azure AI Vision Concepts
3. Design a Vision Solution
4. Responsible AI Review

Lab evidence: You should have a scenario mapping table, Azure AI Vision capability notes, and responsible AI review.

LAB 4 - STEP FLOW

Lab 4 guided steps

- 1. Match Vision Scenarios
- 2. Explore Azure AI Vision Concepts
- 3. Design a Vision Solution
- 4. Responsible AI Review

Lab 4 checkpoint

Checkpoint questions

- What is OCR?
- How is object detection different from image classification?
- Why is face analysis sensitive?
- Which Azure service supports common vision workloads?

Course focus

Vision questions often ask which workload or Azure service fits the scenario.

Natural Language Processing and Azure AI Language

LAB 05

The retail company receives thousands of customer comments. It wants to detect sentiment, extract key topics, identify product names, and route complaints.

Objectives

- Identify NLP workloads.
- Explain sentiment analysis, key phrase extraction, entity recognition, and language modeling.
- Describe Azure AI Language capabilities.
- Review responsible AI concerns for text processing.

Validation: You should have NLP workload mapping, Azure AI Language notes, and a feedback pipeline design.

LAB 5 - DIAGRAM

Lab 5 workflow diagram

1

1. Map NLP Workloads

>

2

2. Review Azure AI Language

>

3

3. Design a Feedback Pipeline

>

4

4. Responsible AI Review

Lab evidence: You should have NLP workload mapping, Azure AI Language notes, and a feedback pipeline design.

Lab 5 guided steps

- 1. Map NLP Workloads
- 2. Review Azure AI Language
- 3. Design a Feedback Pipeline
- 4. Responsible AI Review

Lab 5 checkpoint

Checkpoint questions

- What is sentiment analysis?
- What is entity recognition?
- What is key phrase extraction?
- Why can NLP results require human review?

Course focus

NLP questions often describe text, speech, translation, sentiment, or entity extraction scenarios.

Speech, Translation, Conversational AI

LAB 06

The retail company wants a multilingual customer support experience with voice input, translated text, and a chatbot for common questions.

Objectives

- Describe speech recognition and synthesis.
- Explain translation workloads.
- Recognize conversational AI use cases.
- Map scenarios to Azure AI Speech, Translator, and Bot-related services.

Validation: You should have scenario mappings, Azure service notes, support assistant flow, and responsible AI review.

LAB 6 - DIAGRAM

Lab 6 workflow diagram

1. 1. Match Speech and Translation Scenarios
2. 2. Review Azure Services
3. 3. Design a Support Assistant
4. 4. Responsible AI Review

Lab evidence: You should have scenario mappings, Azure service notes, support assistant flow, and responsible AI review.

LAB 6 - STEP FLOW

Lab 6 guided steps

- 1. Match Speech and Translation Scenarios
- 2. Review Azure Services
- 3. Design a Support Assistant
- 4. Responsible AI Review

Lab 6 checkpoint

Checkpoint questions

- What is speech recognition?
- What is speech synthesis?
- What is translation used for?
- Why is human escalation important in conversational AI?

Course focus

Speech and translation questions usually test whether you can identify the workload and the right Azure AI service.

Document Intelligence and Knowledge Mining

LAB 07

The retail company processes invoices, receipts, and supplier forms manually. It wants to extract structured information and make documents searchable.

Objectives

- Identify document processing workloads.
- Explain form extraction and OCR.
- Describe Azure AI Document Intelligence.
- Explain knowledge mining concepts.

Validation: You should have a document task map, extraction flow, knowledge mining notes, and responsible AI review.

LAB 7 - DIAGRAM

Lab 7 workflow diagram

1

1. Identify Document Tasks

>

2

2. Review Azure AI Document Intelligence

>

3

3. Design an Invoice Processing Flow

>

4

4. Review Knowledge Mining

>

5

5. Responsible AI Review

Lab evidence: You should have a document task map, extraction flow, knowledge mining notes, and responsible AI review.

LAB 7 - STEP FLOW

Lab 7 guided steps

- 1. Identify Document Tasks
- 2. Review Azure AI Document Intelligence
- 3. Design an Invoice Processing Flow
- 4. Review Knowledge Mining
- 5. Responsible AI Review

Lab 7 checkpoint

Checkpoint questions

- What is document processing?
- How is OCR related to document intelligence?
- Why are confidence scores useful?
- What is knowledge mining?

Course focus

Document processing questions may reference OCR, forms, invoices, extraction, and search.

Generative AI, Azure AI Foundry, Azure OpenAI, Model Catalog

LAB 08

The retail company wants to use generative AI to draft product descriptions, summarize customer reviews, and help support agents respond faster.

Objectives

- Explain generative AI concepts.
- Identify common generative AI scenarios.
- Describe Azure AI Foundry, Azure OpenAI, and model catalog capabilities.
- Apply responsible AI considerations for generative AI.

Validation: You should have a generative AI glossary, scenario map, Azure capability notes, and safe assistant design.

Lab 8 workflow diagram

1

1. Define Generative AI Terms

>

2

2. Match Generative AI Scenarios

>

3

3. Review Azure Capabilities

>

4

4. Responsible AI Review

>

5

5. Design a Safe Assistant

Lab evidence: You should have a generative AI glossary, scenario map, Azure capability notes, and safe assistant design.

Lab 8 guided steps

- 1. Define Generative AI Terms
- 2. Match Generative AI Scenarios
- 3. Review Azure Capabilities
- 4. Responsible AI Review
- 5. Design a Safe Assistant

LAB 8 - REVIEW

Lab 8 checkpoint

Checkpoint questions

- What is generative AI?
- What is a prompt?
- What is grounding?
- Why is responsible AI especially important for generative AI?

Course focus

Current AI fundamentals material emphasizes generative AI, Azure AI Foundry, Azure OpenAI, and model catalog concepts.

AI Solution Design, Security, Cost, Governance

LAB 09

The retail company wants one AI solution that analyzes customer feedback, summarizes themes, and routes urgent complaints. You must recommend services and controls.

Objectives

- Design a simple Azure AI solution.
- Identify security, privacy, and governance controls.
- Review cost considerations.
- Choose appropriate Azure AI services.

Validation: You should have a requirements summary, service selection table, security controls, cost controls, and responsible AI controls.

Lab 9 workflow diagram

1 — 1. Confirm Requirements > 2 — 2. Choose Services > 3 — 3. Add Security Controls > 4 — 4. Review Cost Controls > 5 — 5. Add Responsible AI Controls

Lab evidence: You should have a requirements summary, service selection table, security controls, cost controls, and responsible AI controls.

Lab 9 guided steps

- 1. Confirm Requirements
- 2. Choose Services
- 3. Add Security Controls
- 4. Review Cost Controls
- 5. Add Responsible AI Controls

Lab 9 checkpoint

Checkpoint questions

- Why does AI solution design include governance?
- What is RBAC used for?
- Why are budget alerts useful?
- Why is human review important for high-impact outputs?

Course focus

AI solution questions often test choosing the right service and recognizing responsible AI, privacy, and security concerns.

AI-900 Capstone and Course Review

LAB 10

You must brief a manager on which Azure AI services could support the retail company and what risks must be managed.

Objectives

- Review all Azure AI fundamentals topics.
- Build an end-to-end AI service map.
- Practice workload recognition.
- Create a personal review plan.
- Clean up lab resources.

Validation: You should have a service map, responsible AI action list, ML technique table, review log, and cleanup confirmation.

LAB 10 - DIAGRAM

Lab 10 workflow diagram

1

1. Build a Service Map

>

2

2. Review Responsible AI

>

3

3. Review ML Techniques

>

4

4. Build a Review Log

>

5

Validate and review

Lab evidence: You should have a service map, responsible AI action list, ML technique table, review log, and cleanup confirmation.

Lab 10 guided steps

- 1. Build a Service Map
- 2. Review Responsible AI
- 3. Review ML Techniques
- 4. Build a Review Log
- 5. Clean Up Resources
- 6. Prepare a 7-Day Study Plan

Lab 10 checkpoint

Checkpoint questions

- Which AI workload is easiest for you to identify?
- Which Azure service names are most confusing?
- What is the difference between traditional ML and generative AI?
- What responsible AI principle do you need to review most?

Course focus

The goal is to recognize AI workloads, understand foundational concepts, and describe how Azure AI services support common business scenarios.

COURSE SYNTHESIS

The capstone connects workload, service, and control

A useful Azure AI recommendation names the workload, the service, the data boundary, and the responsible AI guardrail.

REVIEW

Final service selection checklist

- What is the input: image, text, speech, document, tabular data, or prompt?
- What output is needed: prediction, label, extracted field, transcript, translation, summary, or answer?
- Which Azure service is the closest fit?
- What privacy, security, human review, and cost controls are required?
- How will the learner validate the result and clean up resources?

ASSESSMENT

Assessment Flow Reminder

1. **TRAQOM digital attendance** >
2. **Assessment digital attendance** >
3. **Sit written / MCQ assessment** >
4. **Submit answers on LMS/TMS** >
5. **Sign Assessment Summary Record**

The TRAQOM QR code and LMS/TMS submission are handled through the LMS/TMS portal.

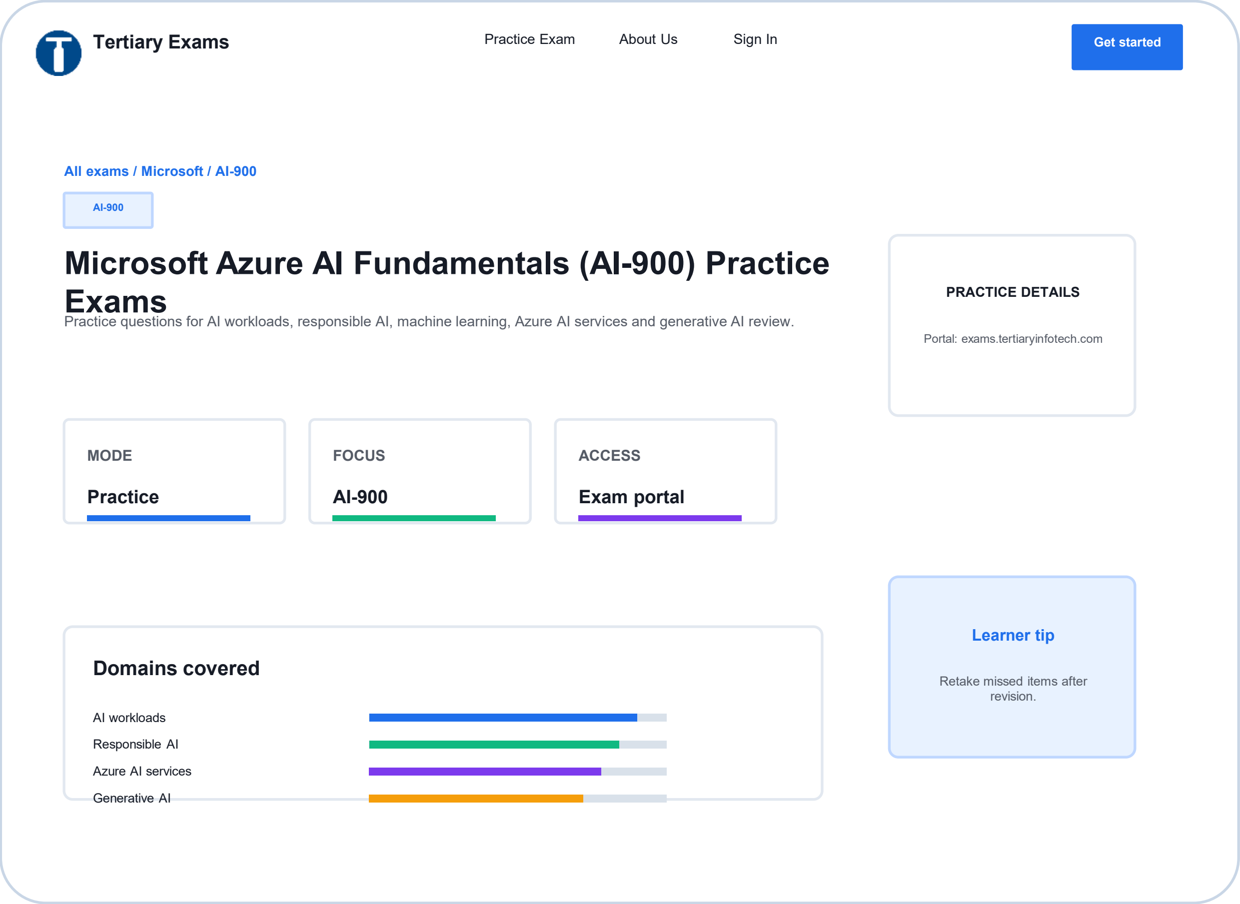
PRACTICE EXAM

Take the Online AI-900 Practice Exams

- Reinforce AI-900 topics with realistic, exam-style questions before assessment day.
- Use practice sets to test AI workloads, responsible AI, machine learning, Azure AI services and generative AI concepts.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from the Tertiary Exams portal, then bring difficult items to the review session.

Access and practise at:

exams.tertiaryinfotech.com



BEFORE YOU GO

Course close

- Complete TRAQOM survey and final attendance.
- Save the Learner Guide and lab notes for post-course review.
- Clean up personal Azure resources or confirm shared cleanup with the trainer.
- Review weak areas using the 7-day study plan in Lab 10.